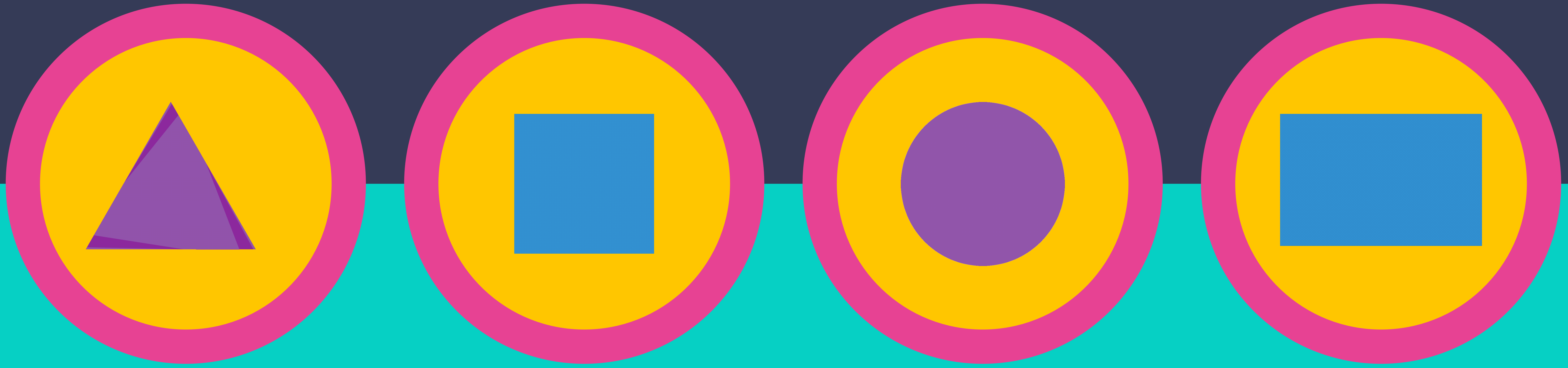
GEOMETRY

Understanding Shapes, Angles, and Dimensions

WHAT IS GEOMETRY?



DEFINITION

- Geometry is a branch of mathematics that studies shapes, sizes, angles, and dimensions of objects.
- It helps us understand the world around us, from the structure of buildings to the patterns in nature.

BASIC CONCEPTS IN GEOMETRY

Introduction to fundamental elements in geometry.

POINTS, LINES, AND PLANES

Point

A location in space with no size (denoted by a dot, like A).

Line

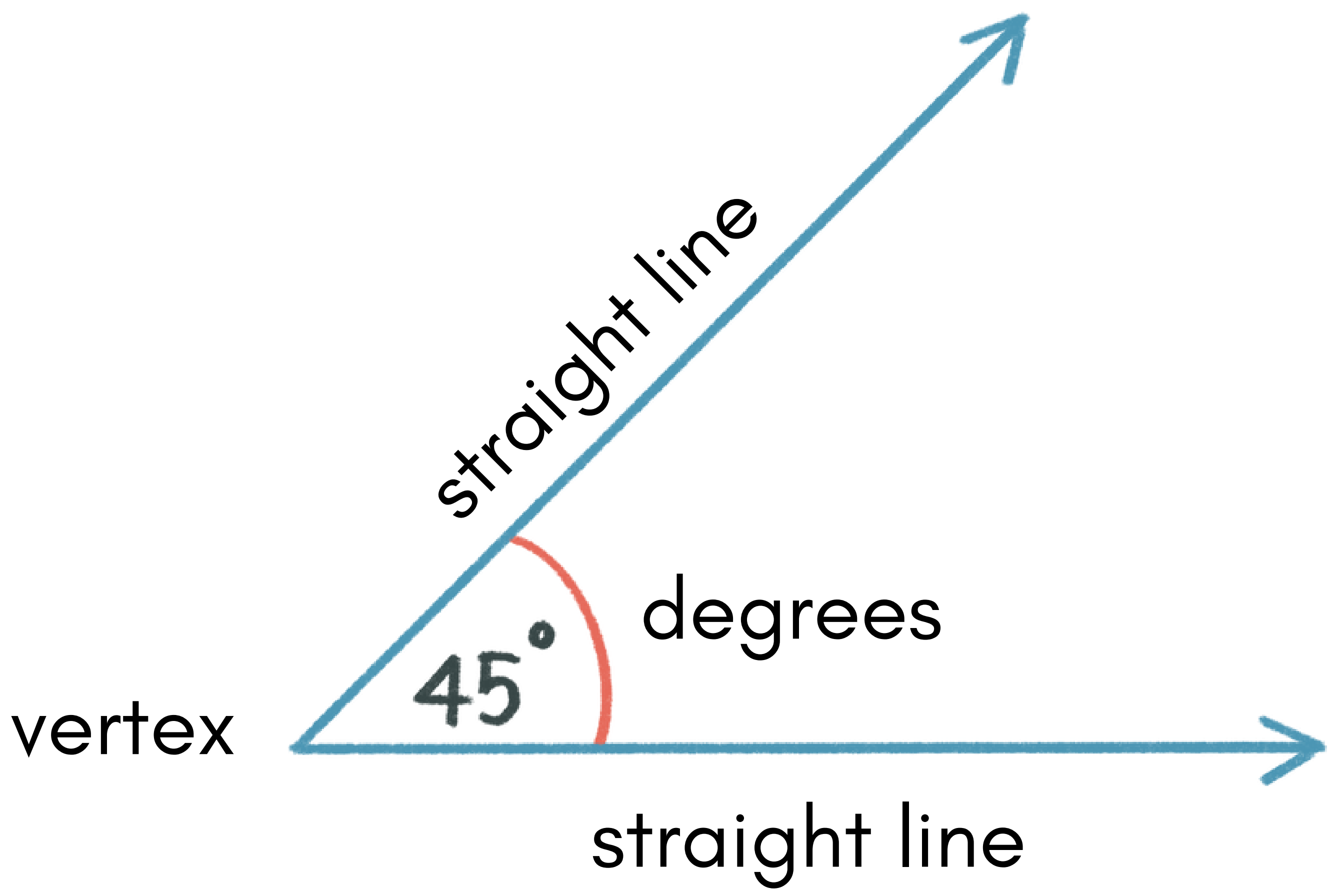
A straight path that extends infinitely in both directions.

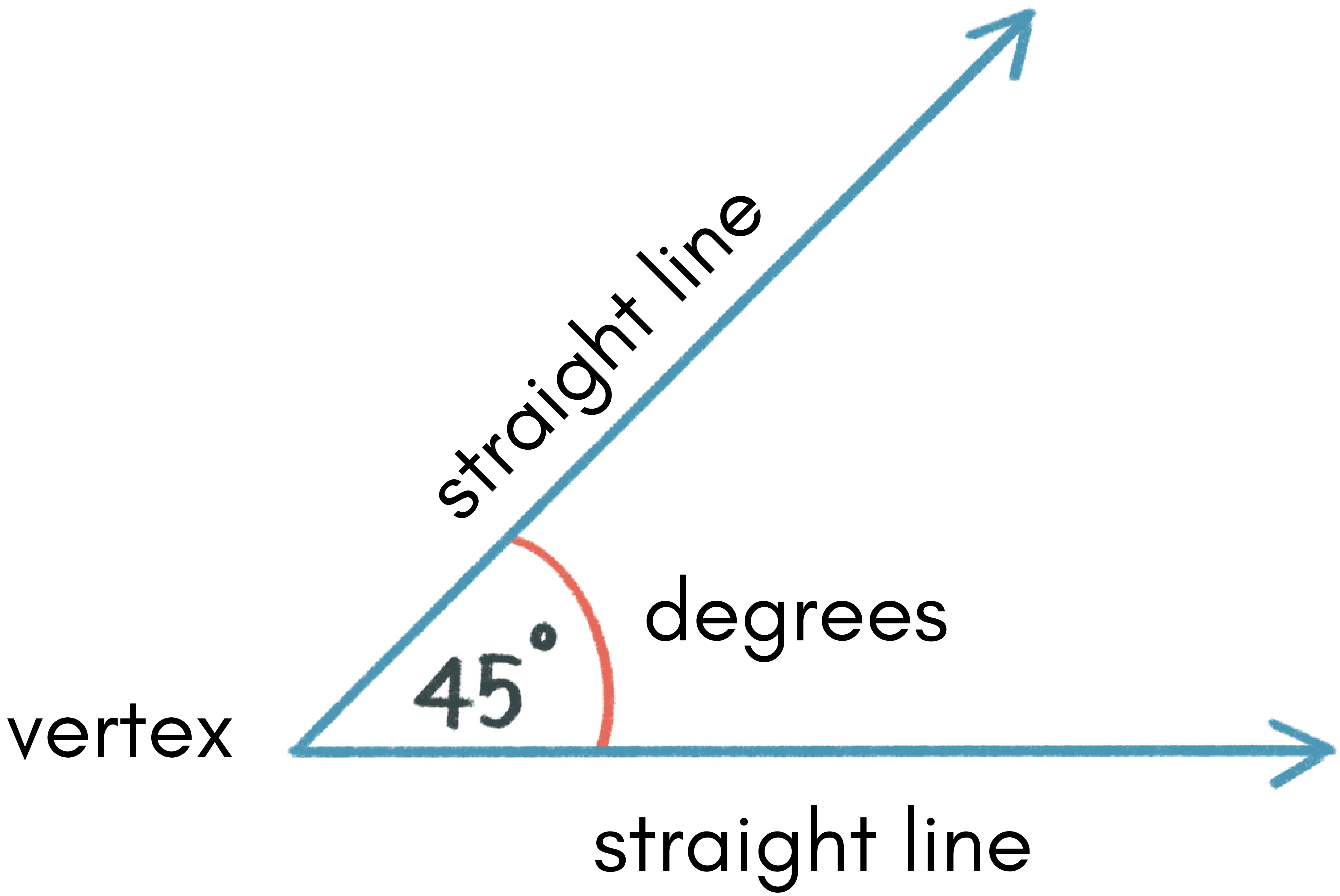
Plane

A flat surface that extends infinitely in all directions.

WHAT ARE ANGLES?

- The space between two intersecting straight lines, at the point in which they meet.
- The point where they meet is called the vertex.
- Angles are measured in degrees.




straight line
degrees
45°
vertex
straight line

TYPES OF ANGLES

smallest

largest

Acute (>0, <90 degrees)

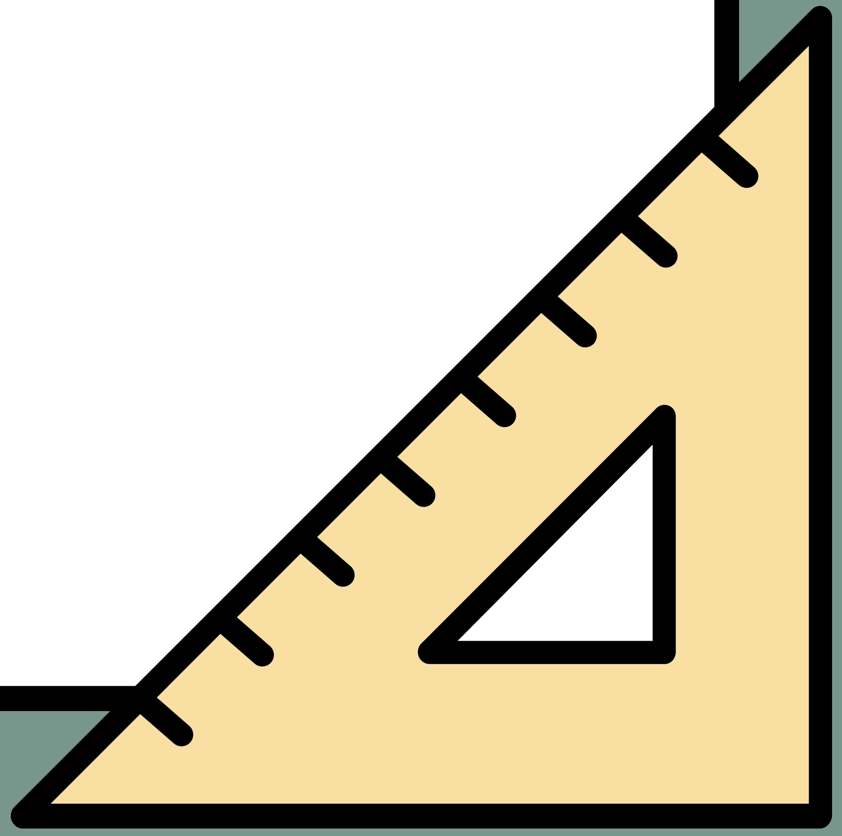
Right (90 degrees)

Obtuse (>90, <180 degrees)

Straight (180 degrees)

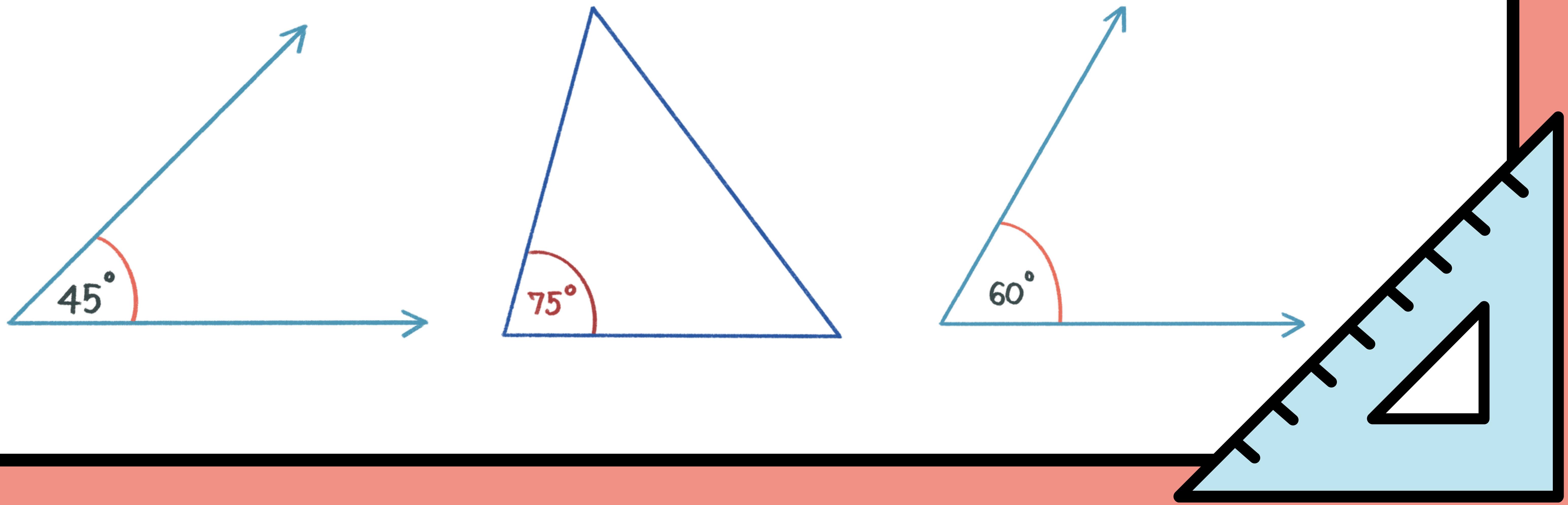
Reflex (>180, <360 degrees)

Revolution (360 degrees)



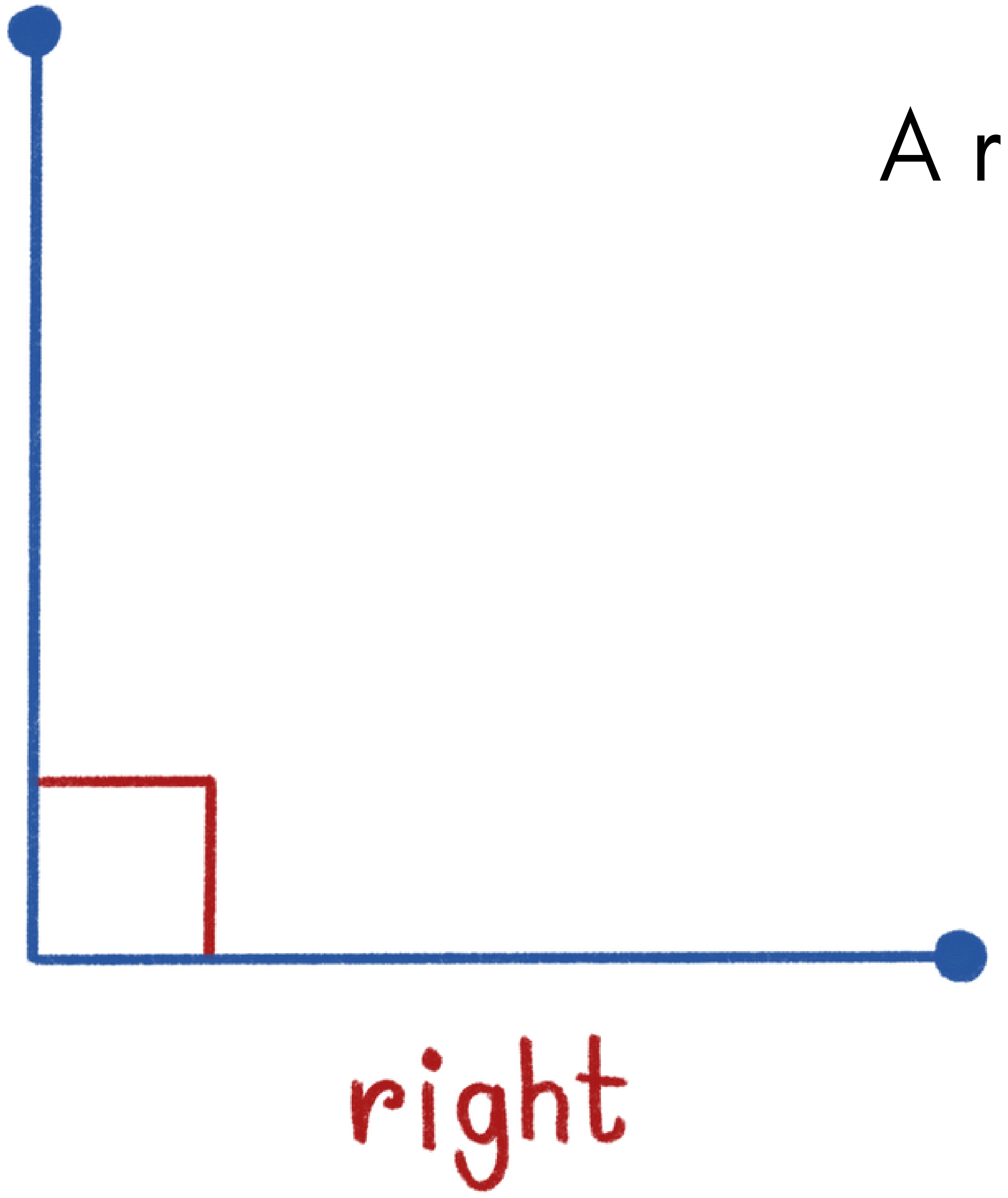
ACUTE ANGLES

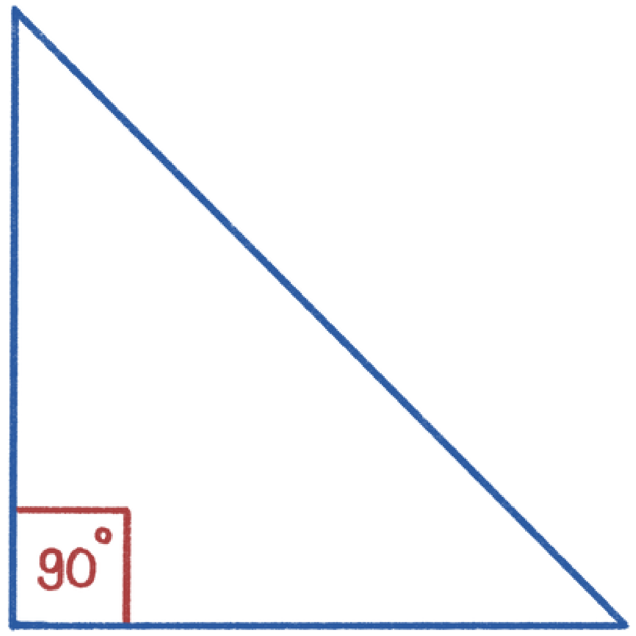
The <u>internal</u> measurement of two straight lines, where the angle measures more than 0° but less than 90°.

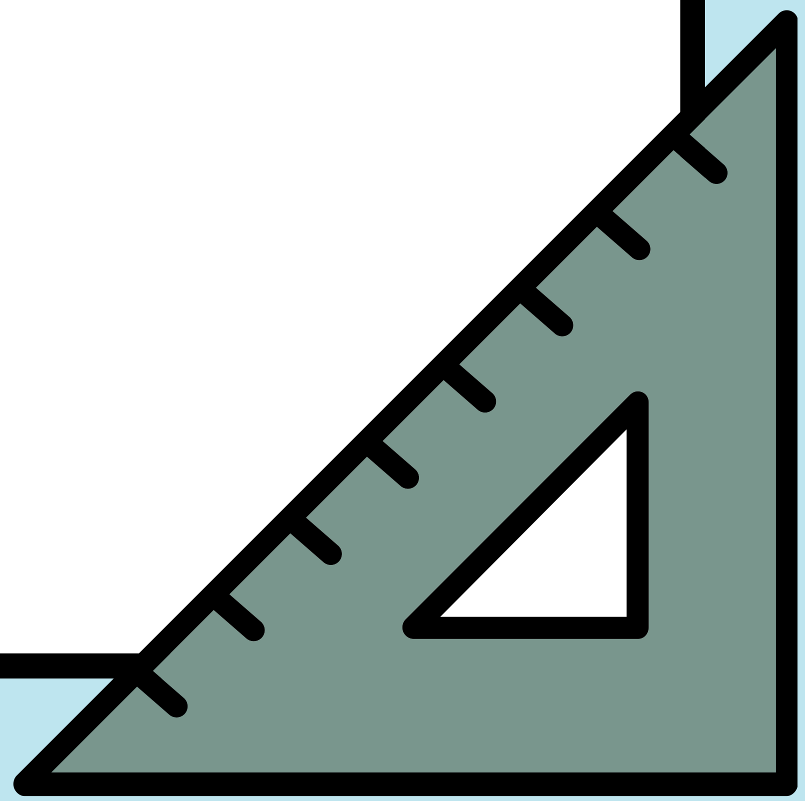


RIGHT ANGLES

A right angle is equal to 90 °

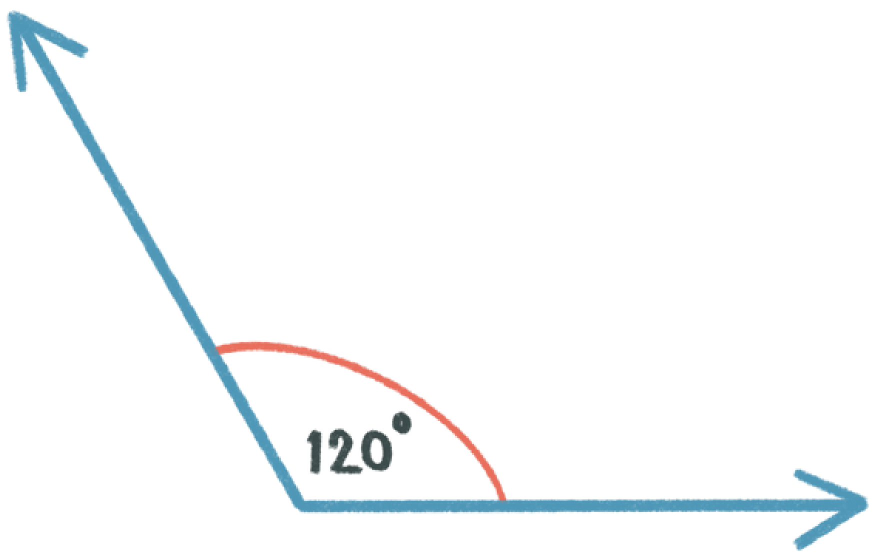


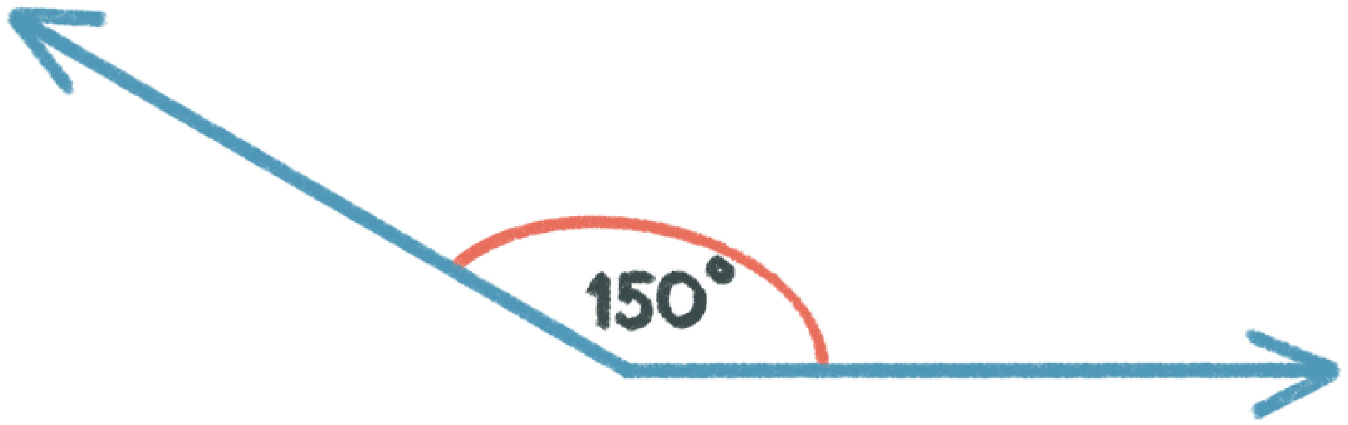




OBTUSE ANGLES

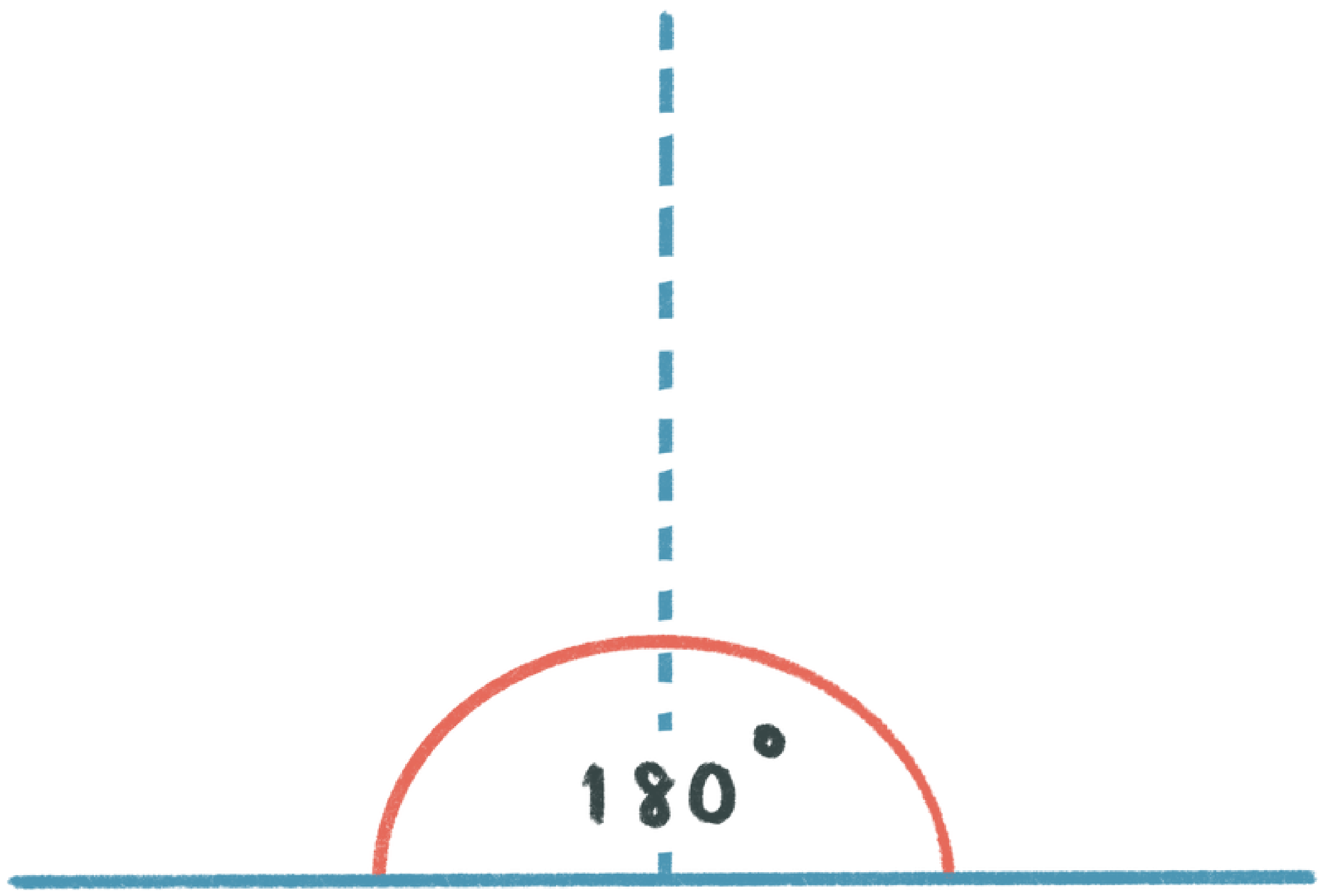
The <u>internal</u> measurement of two straight lines, where the angle measures more than 90° but less than 180°.





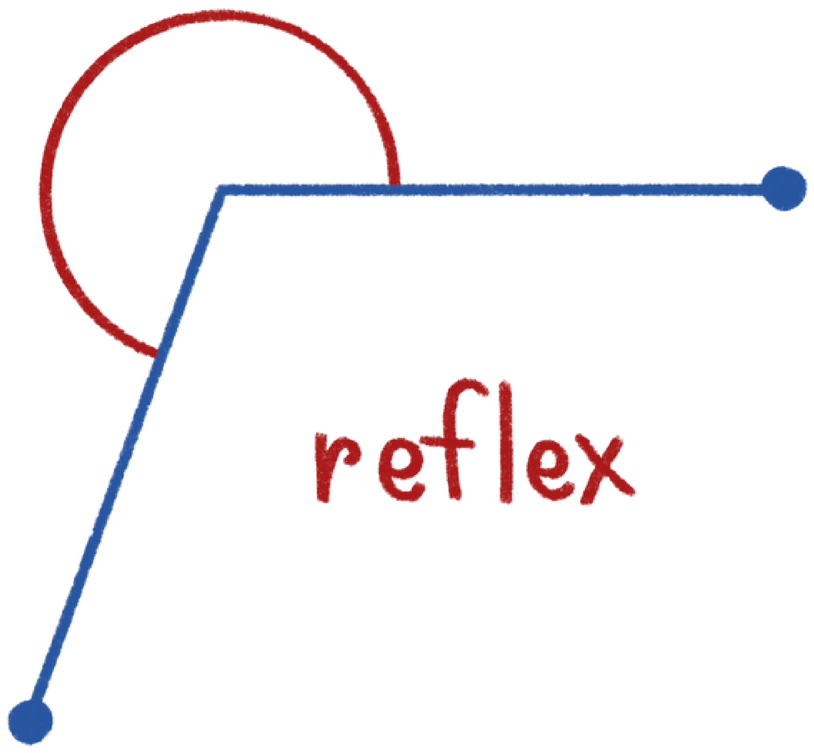
STRAIGHT ANGLES

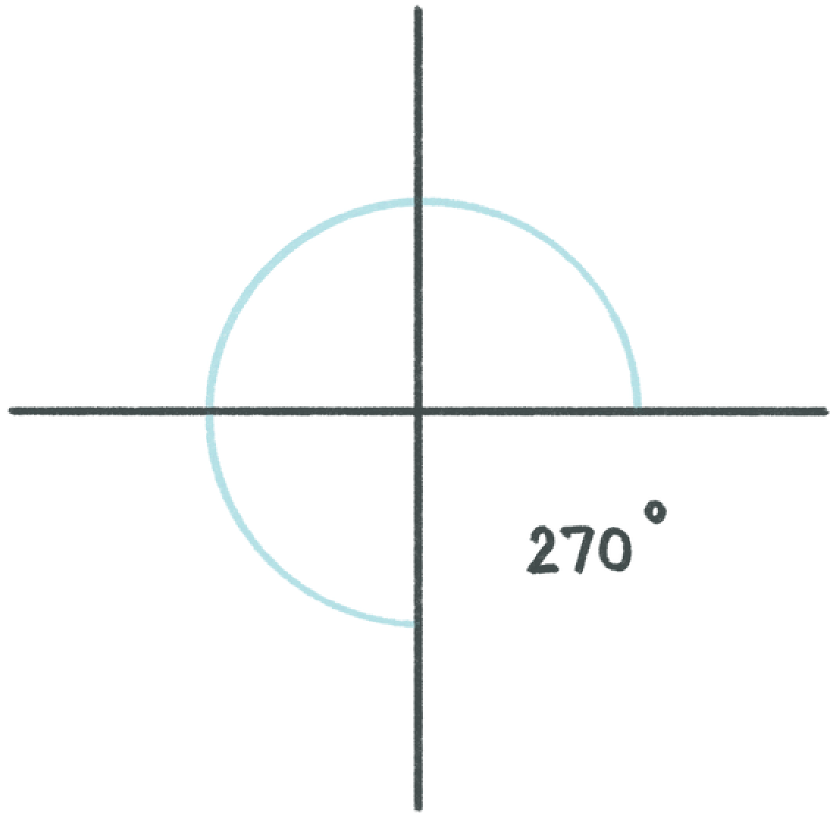
A straight angle is equal to 180 °



REFLEX ANGLES

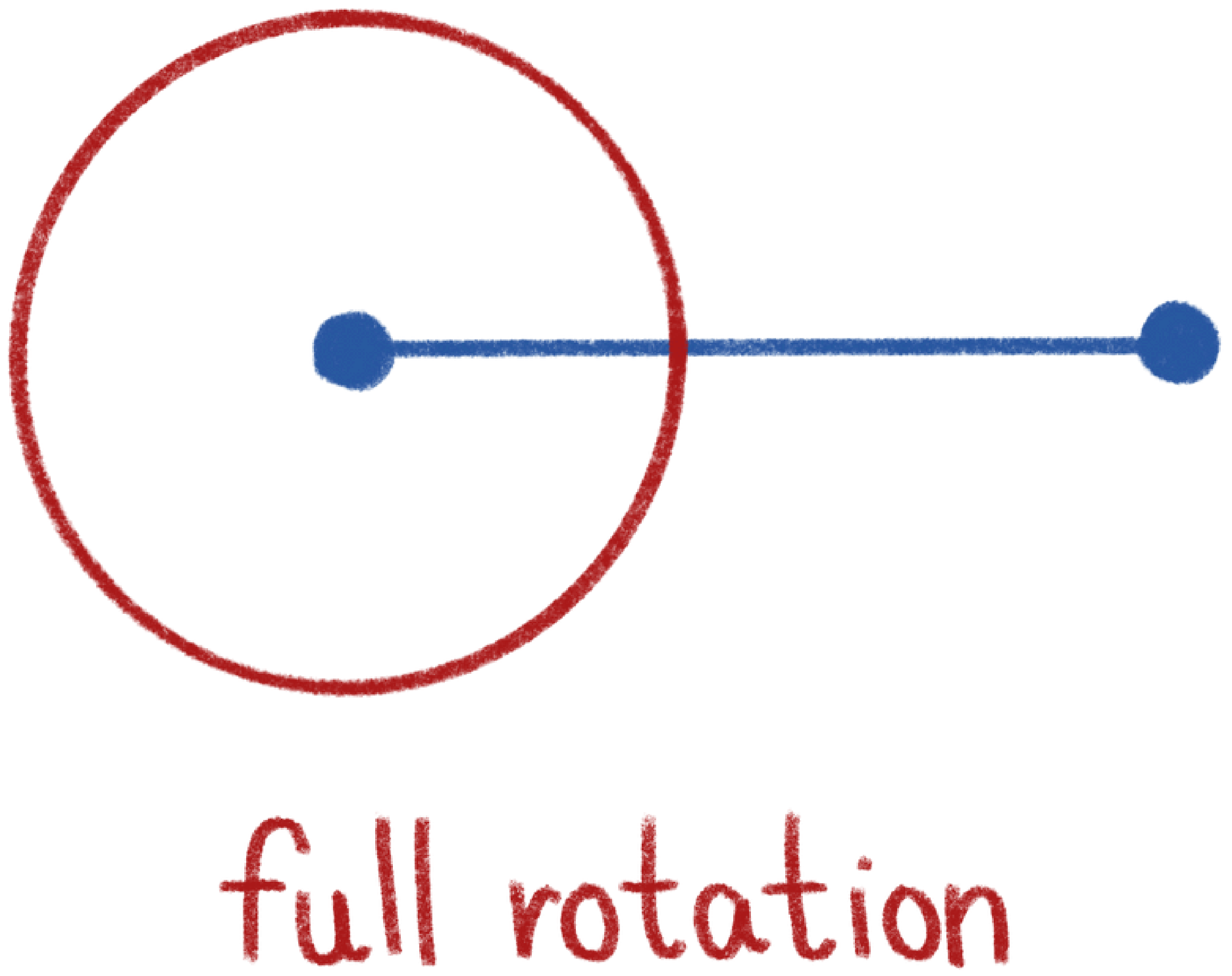
The <u>external</u> measurement of two straight lines, where the angle measures more than 180° but less than 360°.

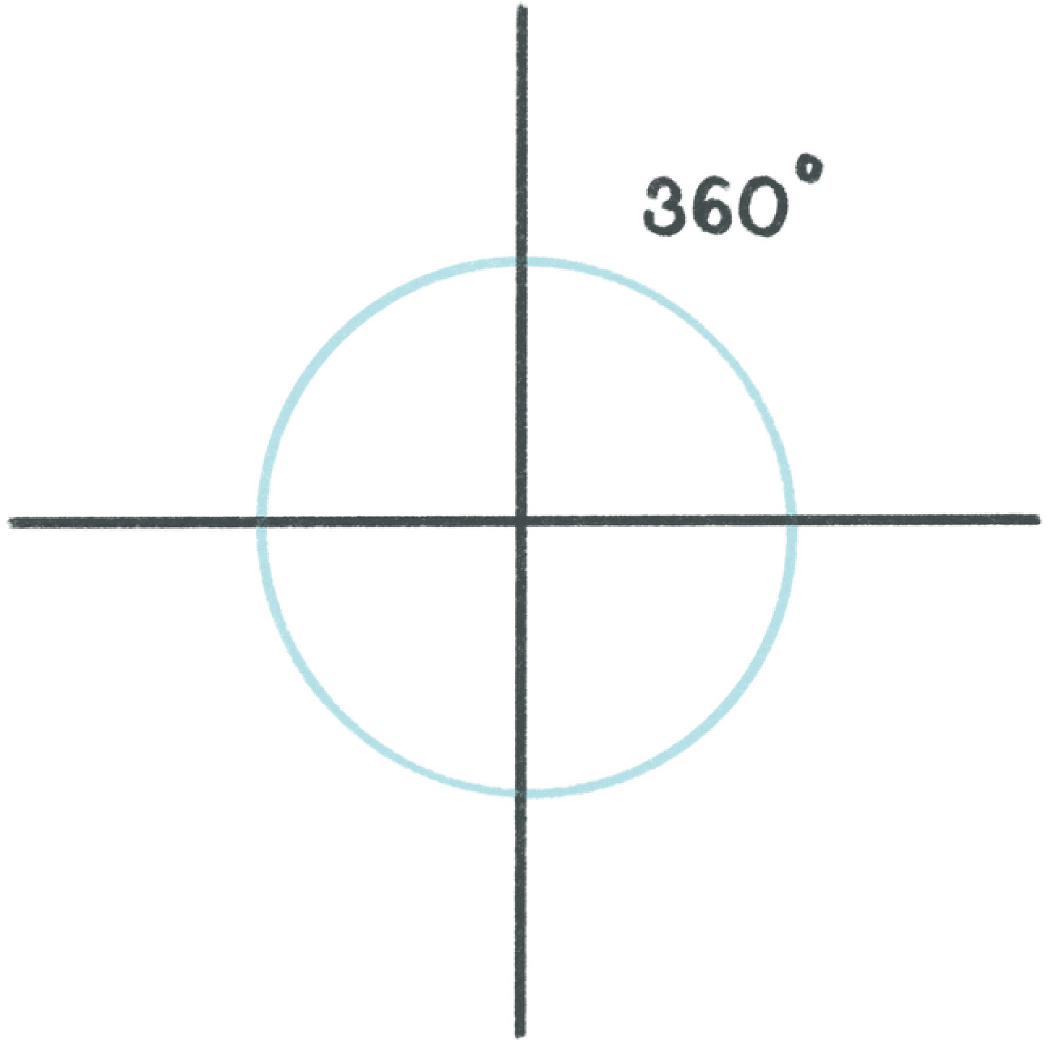




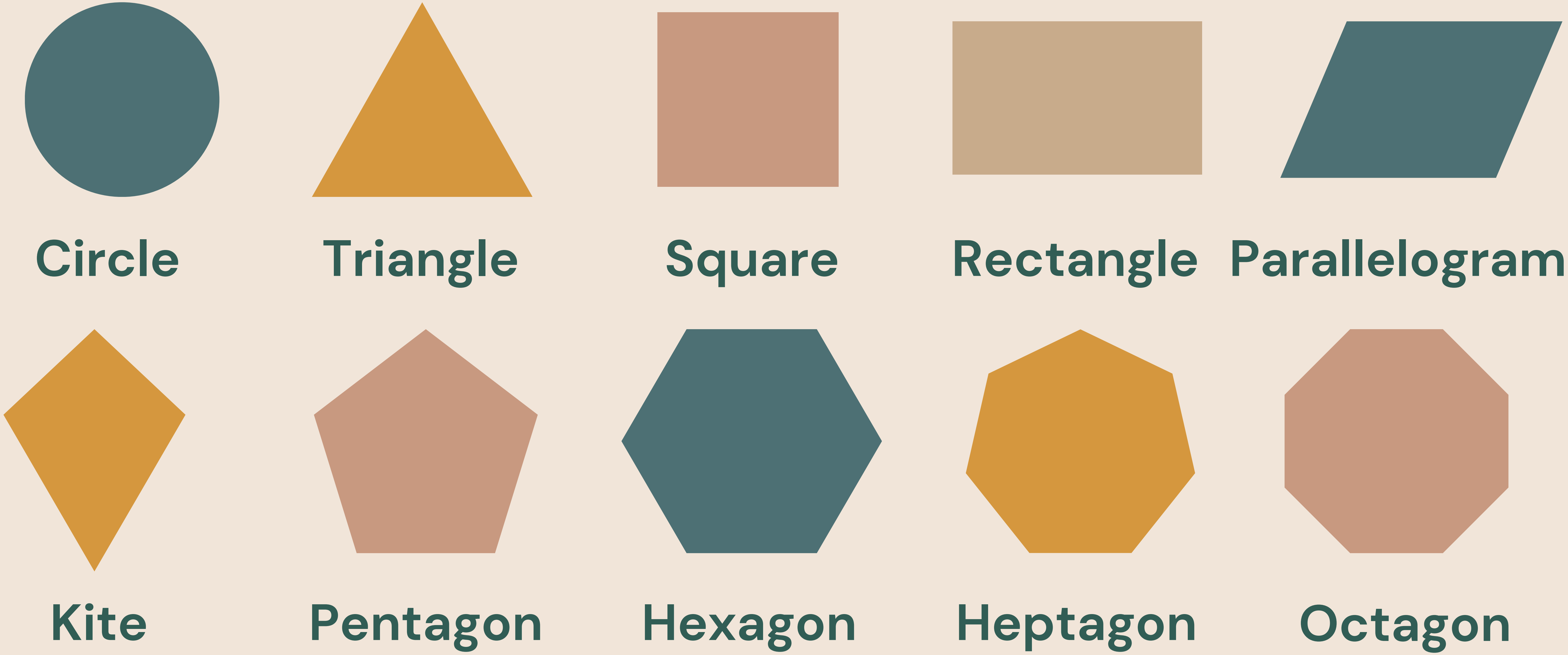
REVOLUTION ANGLES

A full rotation of 360 °





2D SHAPES


Circle
Triangle
Square
Rectangle
Parallelogram
Kite
Pentagon
Hexagon
Heptagon
Octagon

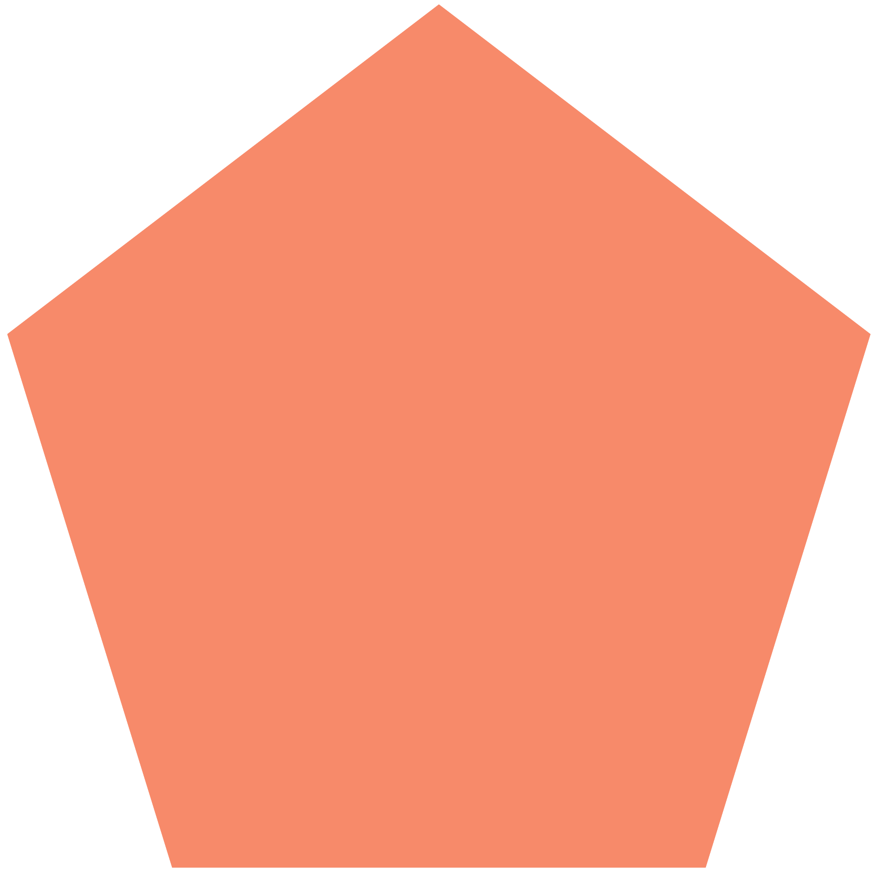
Polygon

A polygon is a closed figure with straight sides.

A regular polygon has all equal side lengths and all equal angle measures.

Regular Polygon

This pentagon is a regular polygon because it has equal sides and equal angles.

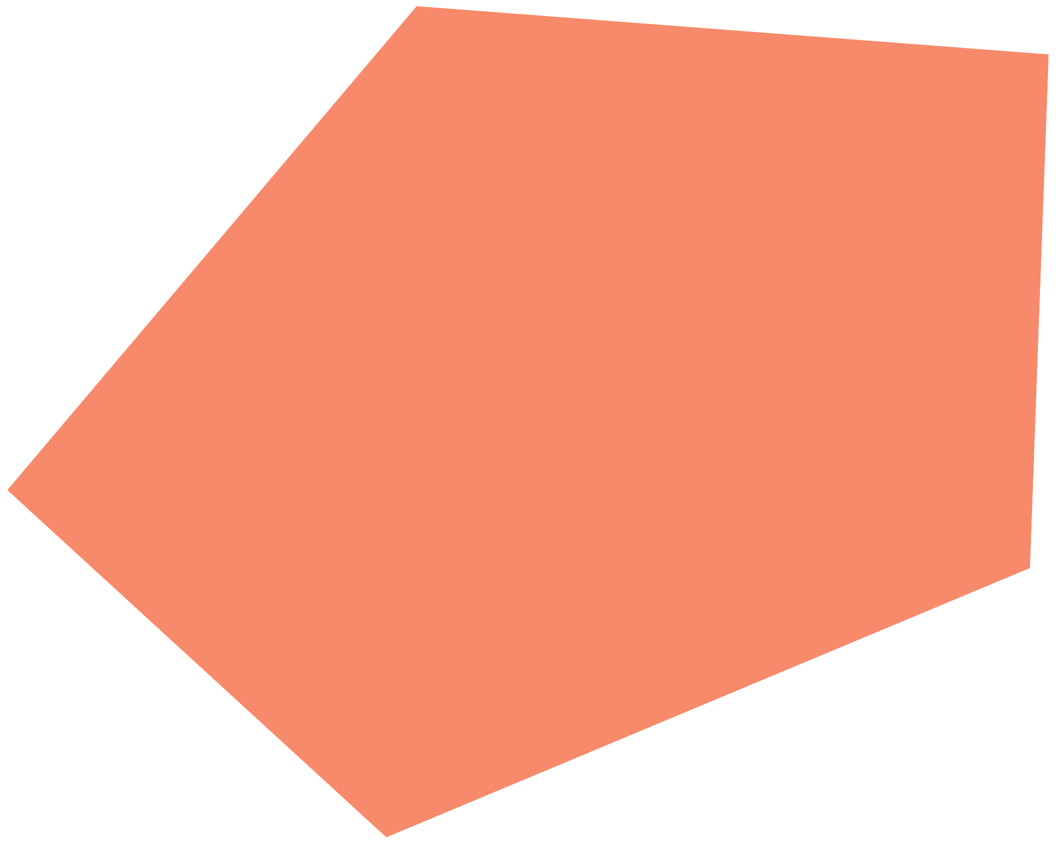


Irregular

If the sides and angles of a polygon are not equal (not the same), the polygon is <u>irregular</u>.

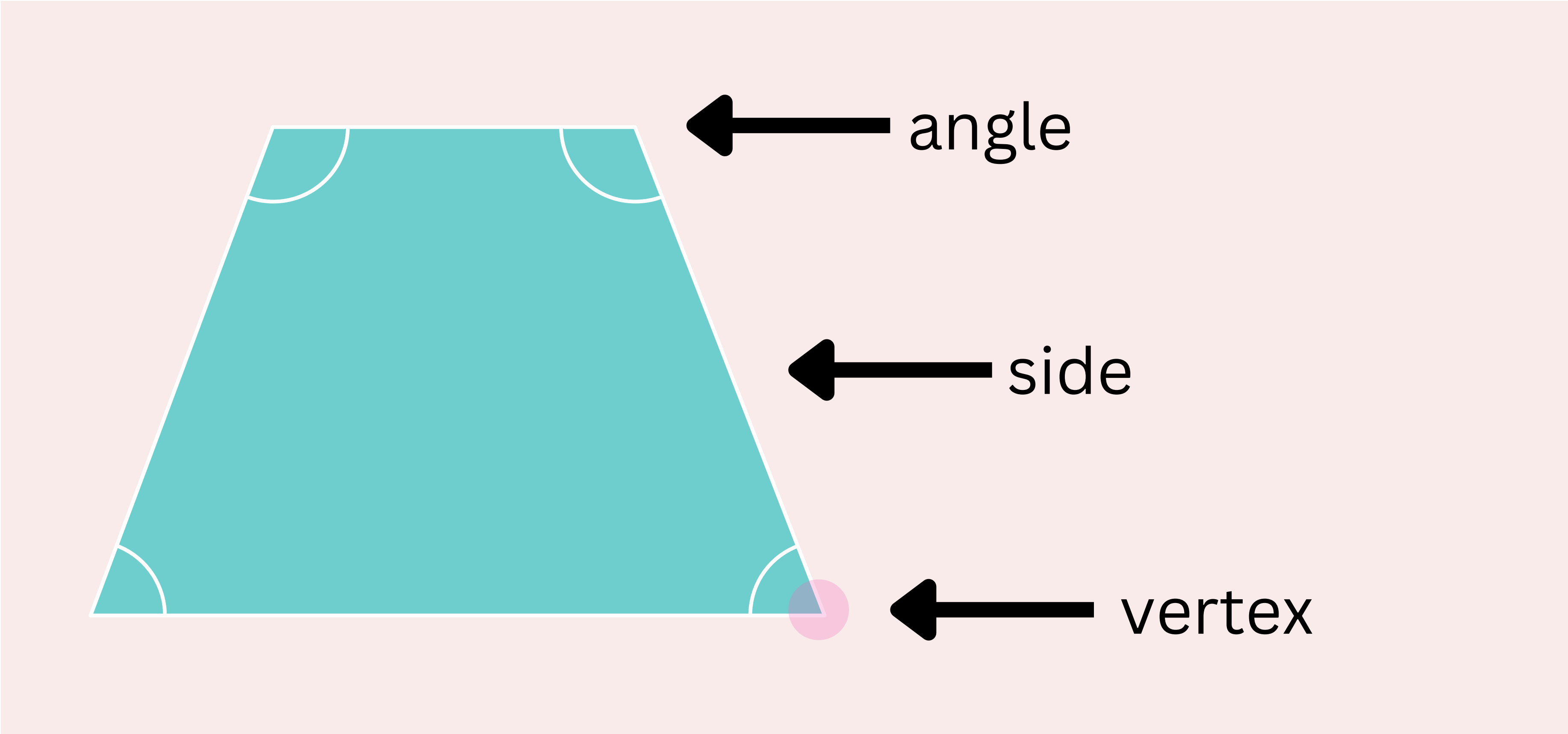
Irregular Polygon

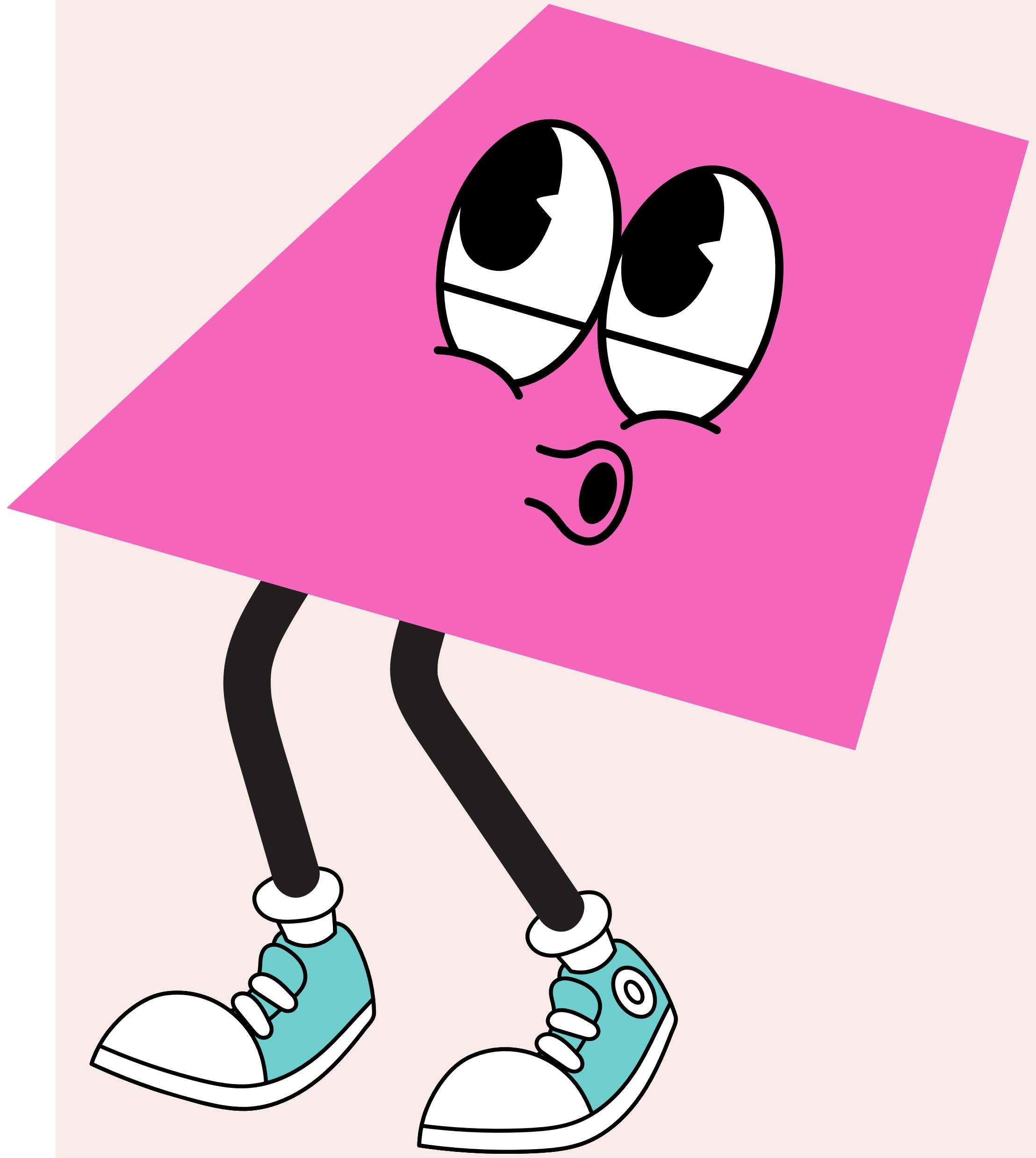
This pentagon is an irregular polygon because it does not have equal sides and equal angles.



FEATURES OF POLYGONS

- **Sides** – edges of the figure.
- **Angles** – where the sides meet.
- **Vertices** – points where sides intersect.

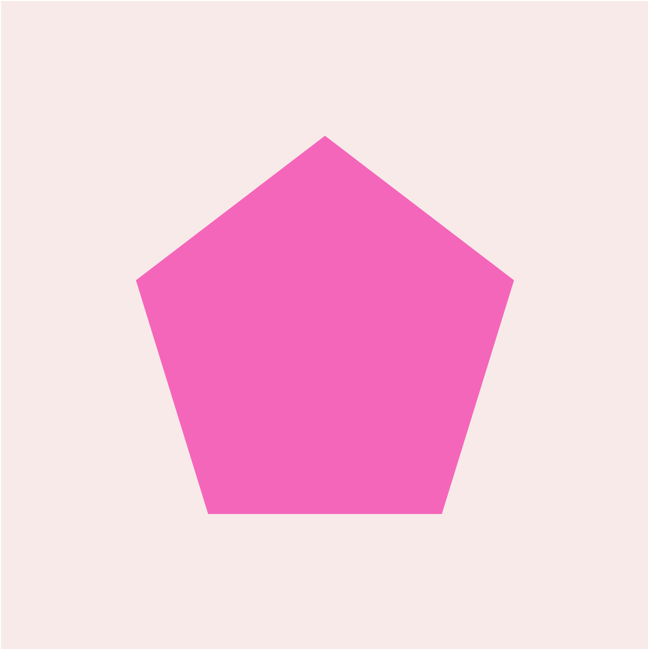




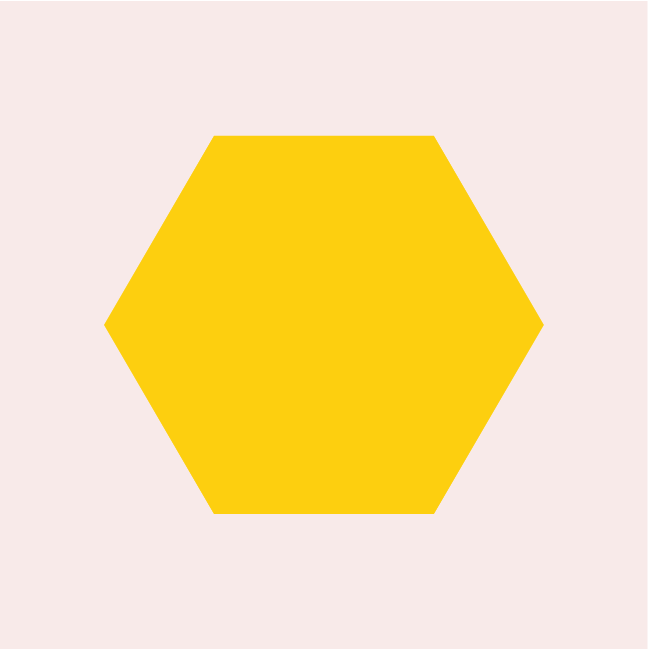
Number of Sides and Names of Polygons

Triangle

Quadrilateral



Pentagon



Hexagon

Count the sides and read the names. Does everything match?

Circle

Number of Sides	1
Vertices	0

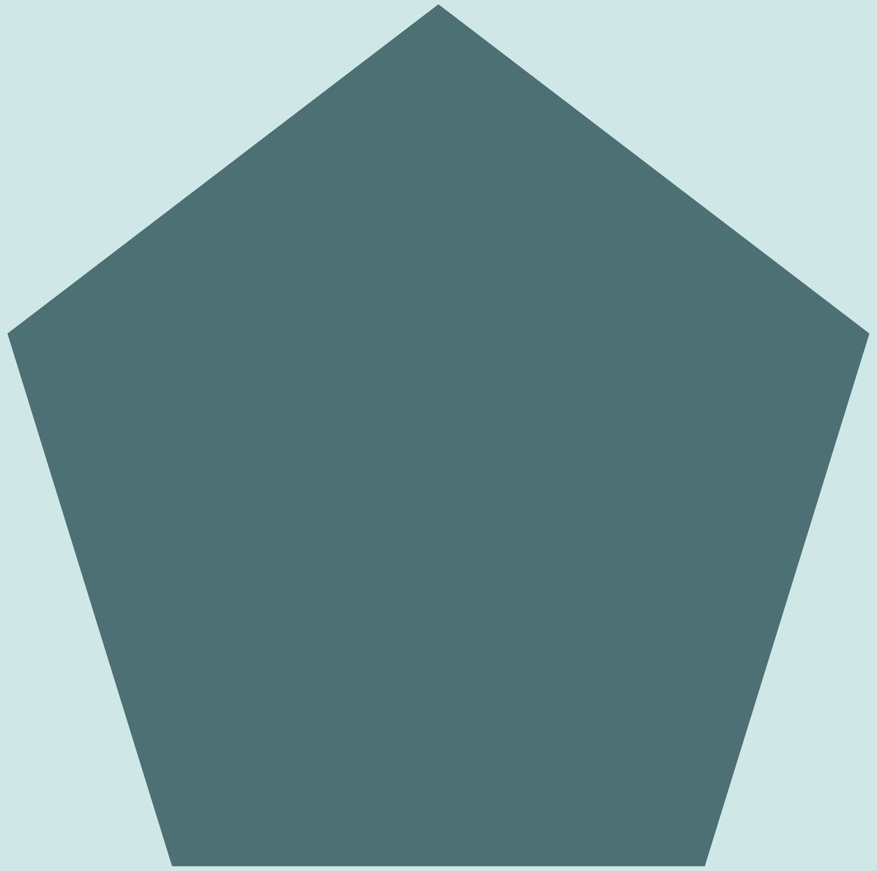
Triangles

Number of Sides	3
Vertices	3

Quadrilaterals

Number of Sides	4
Vertices	4

Pentagon

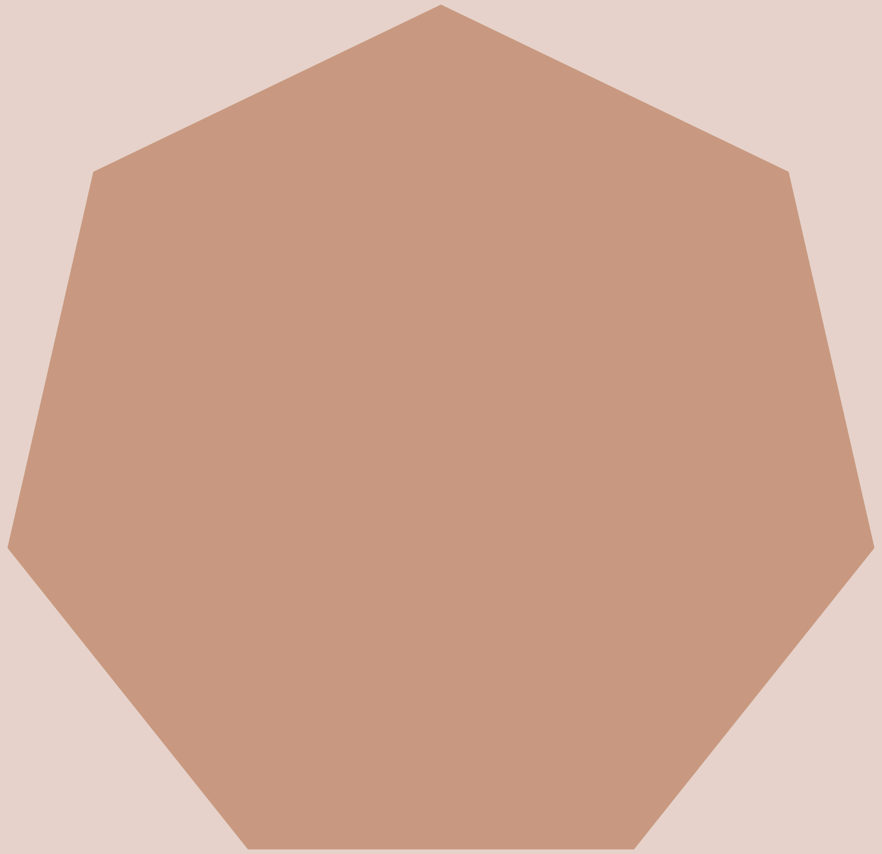


Number of Sides	5
Vertices	5

Hexagon

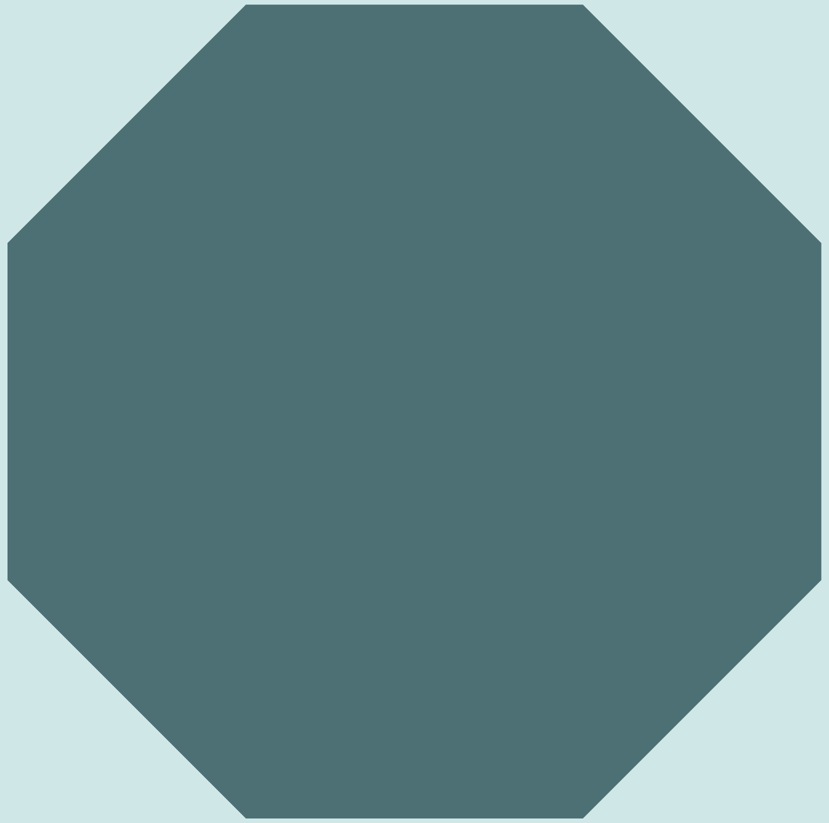
Number of Sides	6
Vertices	6

Heptagon



Number of Sides	7
Vertices	7

Octagon



Number of Sides	8
Vertices	8

REAL-LIFE APPLICATIONS OF GEOMETRY

- Architecture: Used in building designs and structures.
- Engineering: Helps in designing machines and structures.
- Navigation: GPS and maps use geometric concepts.
- Art and Design: Shapes and symmetry are key in creative fields.

CONCLUSION

- Geometry is everywhere!
- It helps us understand and create the world around us.
- Whether in nature, technology, or design, geometry plays a crucial role in shaping our lives.

Grade 4

Stay tuned for part 2

Thank you!